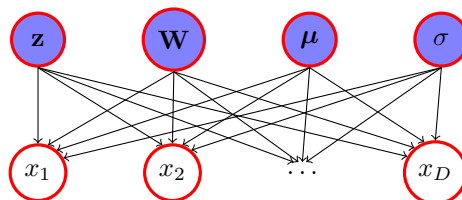


12.6



As per section 8.2.2, the key assumption of the naive Bayes model is that, conditioned on $\mathbf{z}$, $\mathbf{W}$, $\boldsymbol{\mu}$ and σ , the distributions of the input variables $x_1, x_2, \dots, x_D$ are independent.

Every path from any x_i to any other x_j includes one of the four nodes $\mathbf{z}$, $\mathbf{W}$, $\boldsymbol{\mu}$ or σ , and the arrows on the path meet tail-to-tail at those nodes. This implies that all paths between the elements of $\mathbf{x}$ are blocked and, therefore, conditioned on $\mathbf{z}$, $\mathbf{W}$, $\boldsymbol{\mu}$, and σ , the distributions of the input variables $x_1, x_2, \dots, x_D$ are independent.